

ICPC Training Preparation Problem Set

This optional set of 10 CSES problems provides an independent preparation goal for the seven-week ICPC Training Program. Try to solve as many problems as possible. Completing all 10 would indicate strong overall preparation. This is not a graded assignment or formal competition, and progress does not need to be reported.

You may use your preferred programming language and problem-solving approach. Starting with pseudocode is encouraged, and you may ask Alfredo to review a proposed solution.

Using CSES

1. If you do not already have a CSES account, create one at cses.fi/register and log in.
2. Open a linked problem, write a complete program, and save it as one source code file, such as a .py, .cpp, or .java file. The program must read from standard input and write to standard output.
3. On the problem page, choose the matching programming language, upload the source code file, and submit it. Review the result, revise your program if needed, and submit again.

Practice Problems

Problem names in the table are clickable. Attempt the problems before reviewing the self-assessment information on page 3. The problems are presented in random order and do not necessarily follow the week-by-week training sequence.

#	Problem
1	<u>Counting Rooms</u>
2	<u>Factory Machines</u>
3	<u>Book Shop</u>
4	<u>Static Range Sum Queries</u>
5	<u>Course Schedule</u>
6	<u>Message Route</u>
7	<u>Counting Divisors</u>
8	<u>Shortest Routes I</u>
9	<u>Distinct Values Subarrays II</u>
10	<u>Road Reparation</u>

Please attempt the problems on the previous page before continuing.

Then turn to the next page to review the self-assessment guide.

ICPC Training Self-Assessment Guide

A - Independent: I identified the approach and completed the solution independently.

B - Partial: I identified the general approach or wrote pseudocode, but could not complete the implementation.

C - Guided: I recognized the approach only after reviewing the week and topic information below.

D - Unfamiliar: I am still unsure how to approach the problem or the related algorithm is unfamiliar to me.

After attempting the problems, circle one rating for each problem.

#	Problem	Primary Week	Related Topics	Rating
1	Counting Rooms	Week 3	Grid graphs; DFS/BFS; connected components	A / B / C / D
2	Factory Machines	Week 2	Binary search; constraint analysis	A / B / C / D
3	Book Shop	Week 6	Dynamic programming; 0/1 knapsack	A / B / C / D
4	Static Range Sum Queries	Week 1	Constraints; time complexity; prefix sums	A / B / C / D
5	Course Schedule	Week 5	Directed graphs; DAGs; indegrees; topological sorting	A / B / C / D
6	Message Route	Week 4	BFS; unweighted shortest paths; path reconstruction	A / B / C / D
7	Counting Divisors	Week 5	Prime factorization; divisibility; counting divisors	A / B / C / D
8	Shortest Routes I	Week 4	Weighted graphs; priority queues; Dijkstra's algorithm	A / B / C / D
9	Distinct Values Subarrays II	Week 2	Hash maps; frequency counting; two pointers; sliding windows	A / B / C / D
10	Road Reparation	Week 5	Sorting; greedy; DSU; Kruskal; minimum spanning trees	A / B / C / D

A rating reflects preparation in only part of a week's material. It does not demonstrate mastery of every topic or suggest that a student should skip the session. All students are encouraged to attend every session. Week 7 emphasizes algorithm recognition and competition strategy across the full set.